PHY709: 15 October 2026

Maxwell equations, constitutive relations and electromagnetic boundary conditions

Dr. Junaid Ali, COMSATS University Islamabad

Field equations in matter

$$\nabla \cdot \mathbf{D} = \rho_f, \qquad \nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \qquad \nabla \times \mathbf{H} = \mathbf{J}_f + \partial_t \mathbf{D}$$

The sources here are free charge and free current. Polarization and magnetization enter through D and H.

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \qquad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$$

For a linear, local, isotropic, nondispersive material:

$$\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}, \quad \mathbf{J}_{\text{cond}} = \sigma \mathbf{E}$$

Charge conservation

Taking divergence of Ampere-Maxwell law and substituting Gauss's law gives:

$$\partial_t \rho_f + \nabla \cdot \mathbf{J}_f = 0$$

Boundary reference

Declare n from medium 1 to medium 2. For a stationary ordinary interface with finite bulk fields:

$$\hat{\mathbf{n}} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f$$
$$\hat{\mathbf{n}} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0$$
$$\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}$$
$$\hat{\mathbf{n}} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}_f$$

Normal D is continuous only when free sheet charge vanishes. Tangential H is continuous only when free sheet current vanishes. Normal E and tangential B can jump at material interfaces.

Calculation procedure

1. Draw the interface and declare its normal.
2. Resolve each field into normal and tangential components.
3. Apply the appropriate Maxwell jump condition.
4. Insert the constitutive relation for each medium.
5. Check direction, units, source terms and a limiting case.

Preparation for the next class

Read the worked dielectric and current-sheet examples. Reproduce one pillbox and one loop proof without using the final formula. The wave-equation preview assumes a homogeneous, source-free, lossless medium.